1. 由于Kerr黑洞外部无奇性.

从而在外视界上的积分可以变为在无穷远处球面上的积分 $\longrightarrow$ 高斯定理.

从而 $M_H = M_{ADM}$, $J_H = J_{ADM}$.

$M_{ADM} = \frac{1}{16\pi}\int_{S_\infty}(\partial_i h_{ij} - \partial_j h_{ii}) n^j ds$ ，其中 h_{ij} 为三维空间度规与欧氏度规的差.

n^j 为球面外法向量.

在其主要由 $\partial_r h_{tt}$ 贡献. 即 $h_{tt} = g_{tt} - \eta_{tt} \simeq -(1-\frac{2M}{r}) + 1 = \frac{2M}{r}$

$\Rightarrow M_{ADM} = \frac{1}{16\pi}\int \frac{2M}{r^2}\cdot r^2 d\theta d\phi = M$

同理 $J_{ADM} = Ma$

从而: $M_H = M$, $J_H = Ma$

$\because A = \int_{\mathcal{H}} \sqrt{g_{\theta\theta} g_{\phi\phi}}\, d\theta d\phi = 4\pi(r_+^2 + a^2)$

$\kappa = \frac{r_+ - M}{r_+^2 + a^2}$, $\Omega_H = \frac{a}{r_+^2 + a^2}$

$\therefore 2\Omega_H J_H + \frac{\kappa A}{4\pi} = \frac{2a^2 M}{r_+^2 + a^2} + \frac{1}{4\pi}\frac{r_+ - M}{r_+^2 + a^2}\cdot 4\pi(r_+^2 + a^2) = M$

$= \frac{2a^2 M}{2Mr_+} + r_+ - M = \frac{a^2}{r_+} + r_+ - M = 2M - M = M$

2. 类光测地线汇的 Raychaudhuri 方程为: $\frac{d\theta}{dv} = -\frac{1}{2}\theta^2 - \sigma^{\mu\nu}\sigma_{\mu\nu} - R_{\mu\nu}\xi^\mu\xi^\nu$

$\because R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi T_{\mu\nu} \Longrightarrow R_{\mu\nu}\xi^\mu\xi^\nu - \frac{1}{2}Rg_{\mu\nu}\xi^\mu\xi^\nu = 8\pi T_{\mu\nu}\xi^\mu\xi^\nu$ $\quad \because g_{\mu\nu}\xi^\mu\xi^\nu = 0$

$\Rightarrow R_{\mu\nu}\xi^\mu\xi^\nu = 8\pi T_{\mu\nu}\xi^\mu\xi^\nu$

$\Rightarrow \frac{d\theta}{dv} = -\frac{1}{2}\theta^2 - \sigma^{\mu\nu}\sigma_{\mu\nu} - 8\pi T_{\mu\nu}\xi^\mu\xi^\nu$ $\quad \because \theta \ll 1 \Rightarrow \frac{d\theta}{dv} = -\sigma^{\mu\nu}\sigma_{\mu\nu} - 8\pi T_{\mu\nu}\xi^\mu\xi^\nu$

$\frac{dA}{dv} = \oint_{\mathcal{H}(v)} \theta ds$ $\quad$ 且 $\theta = -\frac{1}{\kappa}\frac{d\theta}{dv}$

从而 $\frac{dA}{dv} = \oint_{\mathcal{H}(v)} -\frac{1}{\kappa}(-\sigma^{\mu\nu}\sigma_{\mu\nu} - 8\pi T_{\mu\nu}\xi^\mu\xi^\nu) ds$

$= \frac{8\pi}{\kappa}\oint_{\mathcal{H}(v)} (\frac{1}{8\pi}\sigma^{\mu\nu}\sigma_{\mu\nu} + T_{\mu\nu}\xi^\mu\xi^\nu) ds$

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$$\langle \phi_I', \phi_J' \rangle = \sum_k \sum_l \langle \alpha_{Ik}\phi_k + \beta_{Ik}\phi_k^*, \alpha_{Jl}\phi_l + \beta_{Jl}\phi_l^* \rangle$$
$$= \sum_k \sum_l \alpha_{Ik}^*\alpha_{Jl}\langle \phi_k, \phi_l \rangle + \beta_{Ik}^*\beta_{Jl}\langle \phi_k^*, \phi_l^* \rangle$$
$$= \sum_k \sum_l \delta_{kl}(\alpha_{Ik}^*\alpha_{Jl} - \beta_{Ik}^*\beta_{Jl})$$
$$= \sum_k \alpha_{Ik}^*\alpha_{Jk} - \beta_{Ik}^*\beta_{Jk}$$
$$= (\alpha\alpha^\dagger - \beta\beta^\dagger)_{IJ} = \delta_{IJ} \quad \Rightarrow \alpha\alpha^\dagger - \beta\beta^\dagger = 1$$

$$\langle \phi_I', \phi_J'^* \rangle = \sum_k \sum_l \langle \alpha_{Ik}\phi_k + \beta_{Ik}\phi_k^*, \beta_{Jl}^*\phi_l + \alpha_{Jl}^*\phi_l^* \rangle$$
$$= \sum_k \sum_l \alpha_{Ik}^*\beta_{Jl}^*\langle \phi_k, \phi_l \rangle + \beta_{Ik}^*\alpha_{Jl}^*\langle \phi_k^*, \phi_l^* \rangle$$
$$= \sum_k \sum_l \delta_{kl}(\alpha_{Ik}^*\beta_{Jl}^* - \beta_{Ik}^*\alpha_{Jl}^*)$$
$$= \sum_k \alpha_{Ik}^*\beta_{Jk}^* - \beta_{Ik}^*\alpha_{Jk}^*$$
$$= (\alpha^*\beta^\dagger - \beta^*\alpha^\dagger)_{IJ} = 0 \quad \Rightarrow \alpha\beta^T - \beta\alpha^T = 0$$

同理可得: $\alpha^\dagger\alpha - \beta^T\beta^* = 1$, $\alpha^\dagger\beta - \beta^T\alpha^* = 0$

$b_p^R|^R n, p\rangle = \sqrt{n}\,|^R n-1, p\rangle$; $b_p^{R\dagger}|^R n, p\rangle = \sqrt{n+1}\,|^R n+1, p\rangle$

$a_p^{(1)} \propto b_p^R - k b_p^{L\dagger}$ 其中 $k = e^{-\frac{\pi\sigma}{\kappa}}$

从而: $a_p^{(1)}|0_M\rangle \propto \prod_p (\cosh\phi_p)^{-1} \sum_{n=0}^{\infty} k^n [b_p^R \otimes I^L - k I^R \otimes b_p^{L\dagger}]|^R n, p\rangle|^L n, p\rangle$

$$\propto \prod_p (\cosh\phi_p)^{-1} \sum_{n=0}^{\infty} k^n [\sqrt{n}\,|^R n-1, p\rangle|^L n, p\rangle - \sqrt{n+1}\,k\,|^R n, p\rangle|^L n+1, p\rangle$$

$$\propto \prod_p (\cosh\phi_p)^{-1} [\underbrace{-k|0,p\rangle|1,p\rangle}_{n=0} + \underbrace{k|0,p\rangle|1,p\rangle - \sqrt{2}k^2|1,p\rangle|2,p\rangle}_{n=1} + \underbrace{\sqrt{2}k^2|1,p\rangle|2,p\rangle + \cdots}_{n=2}]$$

$$= 0$$

$a_p^{(1)}$ 同理也可湮灭 $|0_M\rangle$. 从而 $|0_M\rangle$ 为闵氏真空.